

Here's a small glimpse of what all has been happening in this field

- Intel's Loihi is a neuromorphic AI chip that can implement spiking neural networks (SNNs). Quite recently, Intel announced the next generation of this chip, Loihi 2.
- Accenture, Airbus, GE, and Hitachi have joined forces with the Intel Neuromorphic Research Community (INRC).
- Samsung, along with researchers from Harvard, published a paper titled 'Neuromorphic electronics based on copying and pasting the brain' in Nature Electronics in September 2021. It presented a new way of reverse-engineering how our brain works.
- The US Air Force and IBM are working together to build a neuromorphic supercomputer, probably based on the next generation of IBM's True North chip.
- A team of Chinese researchers has developed an AI chip, called Tianjic, and deployed it in an autonomous cycle.
- TCS collaborated with BrainChip in 2019 for a demonstration on hand gesture recognition featuring the Akida neuromorphic technology.
- IIT Kanpur's NeuroCHaSe research group is actively working on the development of neuromorphic hardware platforms and ensuring their security against adversary attacks. They are also going to tape out (fabricate) a 3D-neuromorphic computing IC soon.
- MemryX Inc., a startup at the University of Michigan, is developing a programmable AI accelerator chip that promises unmatched performance at relatively low power for high-performance applications.

Another major event that occurred this year was when Mercedes implemented BrainChip's Akida technology in its Vision EQXX electric vehicle. This is definitely a big deal since the Akida technology is tried and tested for a smart automotive experience. All features that the Akida provides, including facial recognition, keyword spotting, etc consume extremely low power.

"This is where we get excited. You'll see a lot of these functionalities in vehicles—recognition of voices, faces, and individuals in the vehicle. This allows the vehicles to have customisation and device personalisation according to the drivers or the passengers as well," says Telson. These really are exciting times.

Neuromorphic hardware, neural networks, and AI

The process in which neurons work is eerily similar to an electric process. Neurons communicate with each other via synapses. Whenever they receive input, they produce electrical signals called spikes (also called action potentials), and the event is called neuron spiking.

When this happens, chemicals called neurotransmitters are released into hundreds of synapses and activate the respective neurons. That's the reason why this process is super-fast.

Artificial neural networks mimic the logic of the human brain, but on a regular computer. The thing is, regular computers work on the Von Neumann architecture, which is extremely different from the architecture of our brain and is very power-hungry. We may not be able to deploy CMOS logic on the Von Neumann architecture for long. We will eventually reach a threshold to which we can exploit silicon. We are

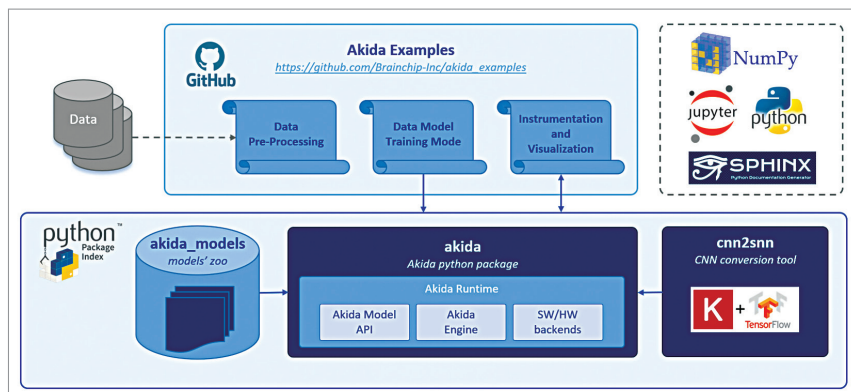
nearing the end of Moore's Law and there is a need to establish a better computing mechanism. Neuro-morphic computing is the solution because neuromorphic hardware realises the structure of the brain in the solid-state.

As we make progress in neuro-morphic hardware, we will be able to deploy neural networks on it. Spiking Neural Network (SNN) is a type of artificial neural network that uses time in its model. It transmits information only when triggered—or, in other words, spiked. SNNs used along with neuromorphic chips will transform the way we compute, which is why they are so important for AI.

How to get started with SNNs

Since the entire architecture of neuromorphic AI chips is different, it is only natural to expect the corresponding software framework to be different too. Developers need to be educated on working with SNNs. However, that is not the case with MetaTF, a free software development framework environment that BrainChip launched in April 2021.

"We want to make our environment extremely simple to use. Having people learn a new development environment is not an efficient way to move forward," says Telson. "We had over 4600 unique users start looking at and playing with MetaTF in 2021. There's a community out there that wants to learn."



Akida MetaTF ML Framework (Source: MetaTF)